

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

INTELLIGENT NOSQL DATABASES

Project Report

Wine Quality Prediction Using Machine Learning and MongoDB

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ABSTRACT

Wine quality assessment is an important task in the food and beverage industry. Traditional wine quality evaluation often involves laboratory testing and sensory assessment by trained individuals. Machine learning provides an opportunity to estimate wine quality from measurable physicochemical properties.

This project, *Wine Quality Prediction Using Machine Learning and MongoDB*, develops a system to predict wine quality using the physicochemical characteristics of red and white wine. The project uses the UCI Wine Quality dataset, containing 1,599 red wine records and 4,898 white wine records.

Random Forest algorithms are implemented for two machine learning tasks: regression and binary classification. The regression models estimate numerical wine quality scores, while the classification models categorize wines as Good or Not Good. Wines with a quality score of 7 or higher are classified as Good.

The system uses Python, Pandas, NumPy, and Scikit-learn for data processing and model development. MongoDB stores the datasets, model evaluation metrics, and prediction history. An interactive dashboard developed using Streamlit enables users to enter wine characteristics, obtain predictions, review model performance, and examine previous predictions.

The regression models achieved R^2 scores of 0.5319 for red wine and 0.5541 for white wine. The corresponding MAE values were 0.4251 and 0.4172. The classification models achieved accuracies of 90.31% and 87.65%, respectively.

The project demonstrates how machine learning, database management, and interactive visualization can be integrated into a practical wine quality prediction application.

Keywords: Machine Learning, Wine Quality Prediction, Random Forest, Regression, Classification, MongoDB, Python, Streamlit.

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1. INTRODUCTION

1.1 Background

Machine learning is a branch of artificial intelligence that enables computers to identify patterns in data and make predictions based on those patterns. It is widely used in areas such as healthcare, finance, manufacturing, agriculture, and quality control.

Wine quality depends on several physicochemical properties, including acidity, residual sugar, chlorides, density, pH, sulphates, and alcohol content. These characteristics can be measured and analyzed to identify relationships with wine quality scores.

Predicting wine quality using machine learning provides a practical example of applying data analysis and predictive modeling to a real-world problem.

1.2 Problem Statement

Evaluating wine quality can require specialized testing and expert judgment. It is useful to develop a system that estimates wine quality from measurable chemical properties.

The problem addressed in this project is to build a machine learning application that:

- Predicts numerical wine quality scores.
- Classifies wine as Good or Not Good.
- Evaluates model performance using multiple metrics.
- Stores datasets, metrics, and prediction history in a database.
- Presents predictions and evaluation results through an interactive dashboard.

1.3 Objectives

The main objectives are:

1. Collect and load red and white wine datasets.
2. Explore and preprocess the available data.
3. Train Random Forest regression and classification models.
4. Evaluate regression models using R^2 score and MAE.
5. Evaluate classification models using Accuracy, Precision, Recall, and F1-score.
6. Integrate MongoDB for persistent data storage.
7. Develop a Streamlit dashboard for predictions and performance analysis.
8. Store and display historical prediction records.

1.4 Scope

The project focuses on predicting wine quality from 11 physicochemical features using the supplied red and white wine datasets.

It includes model training, evaluation, prediction, database integration, and visualization. The current implementation runs locally and uses a local MongoDB server.

The project does not replace laboratory testing or professional wine assessment. Its predictions are estimates based on patterns learned from the available dataset.

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2. LITERATURE REVIEW

Machine learning techniques are frequently used for classification and regression problems involving structured datasets.

2.1 Wine Quality Dataset

The UCI Wine Quality dataset contains physicochemical measurements and quality scores for red and white Portuguese wines. It is commonly used to study relationships between chemical properties and wine quality.

The dataset provides a suitable basis for comparing regression and classification techniques.

2.2 Random Forest

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees.

For regression, it combines the outputs of multiple trees to estimate a numerical value. For classification, it combines the predictions of individual trees to determine a class.

Random Forest is suitable for this project because it can model nonlinear relationships and interactions between input features.

2.3 Regression and Classification

Regression estimates continuous numerical values. In this project, regression predicts the wine quality score.

Classification assigns observations to categories. Here, the classification task uses a threshold of 7 to separate wines into Good and Not Good categories.

2.4 Database Integration

A database provides persistent storage for datasets, evaluation metrics, and prediction records. MongoDB is used because it stores documents in a flexible, JSON-like structure.

2.5 Identified Need

The project combines prediction, evaluation, database storage, and an interactive interface into one application. This allows users to explore predictions and model results without manually running every analysis step.

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3. DATASET DESCRIPTION

3.1 Dataset Source

The project uses the UCI Wine Quality dataset, which contains separate datasets for red and white wine.

The source files used in the implementation are:

- winequality-red.csv
- winequality-white.csv
- winequality.names

3.2 Dataset Size

Dataset	Number of Records
Red wine	1,599
White wine	4,898
Total	6,497

3.3 Input Features

The models use the following 11 physicochemical features:

No.	Feature	Description
1	Fixed acidity	Nonvolatile acids present in wine
2	Volatile acidity	Volatile acids that can affect wine aroma and taste
3	Citric acid	Citric acid concentration
4	Residual sugar	Sugar remaining after fermentation
5	Chlorides	Chloride concentration
6	Free sulfur dioxide	Free sulfur dioxide concentration
7	Total sulfur dioxide	Total sulfur dioxide concentration
8	Density	Density of the wine
9	pH	Measure related to acidity
10	Sulphates	Sulphate concentration
11	Alcohol	Alcohol content

The original quality score is used as the target for regression.

3.4 Classification Target

The numerical quality score is converted into a binary target:

- **Good:** Quality score greater than or equal to 7.
- **Not Good:** Quality score below 7.

This threshold converts the original quality prediction problem into a binary classification task.

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4. SYSTEM REQUIREMENTS

4.1 Hardware Requirements

The project can run on a standard personal computer capable of running Python, MongoDB, and a web browser.

Recommended environment:

- Modern dual-core or better processor.
- At least 8 GB RAM recommended.
- Sufficient storage for the datasets, virtual environment, trained models, and result files.
- Internet access for initial package installation, if dependencies are not already installed.

These are general recommendations rather than measured minimum hardware requirements.

4.2 Software Requirements

Software or Library	Purpose
Python	Main programming language
Pandas	Data manipulation
NumPy	Numerical operations
Scikit-learn	Machine learning and model evaluation
Matplotlib	Plot generation
Seaborn	Statistical visualization
MongoDB	Database storage
PyMongo	Python–MongoDB integration
Streamlit	Interactive web dashboard
Joblib	Saving and loading trained models

The project was developed in a Python 3.13.4 virtual environment.

5. SYSTEM ARCHITECTURE

The system consists of five main components.

5.1 Data Layer

The red and white wine CSV files provide the input data. The data loading script imports the records into MongoDB.

5.2 Processing Layer

Python and Pandas handle data loading and preparation. The models use the physicochemical features to learn relationships with wine quality.

5.3 Machine Learning Layer

Four trained models are used:

- Red wine regression model.
- Red wine classification model.
- White wine regression model.
- White wine classification model.

The trained models are saved using Joblib for later use.

5.4 Database Layer

MongoDB stores wine records, model metrics, and predictions.

5.5 Presentation Layer

Streamlit provides the user interface. Users can enter wine characteristics, view predictions, inspect model performance, and browse prediction history.

System Workflow

CSV datasets → Data loading → MongoDB storage → Model training → Model evaluation → Saved models → Streamlit dashboard → Prediction and history storage.

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6. METHODOLOGY

6.1 Data Loading

The data loading script reads the CSV files using Pandas. Since the original dataset uses semicolons as separators, the files are read with the appropriate delimiter.

The records are inserted into the corresponding MongoDB collections.

6.2 Data Preparation

The physicochemical features are used as model inputs, and the quality score is used as the target.

The data is divided into training and testing sets using an 80:20 split, with random_state=42.

This produces separate data for model training and evaluation.

6.3 Regression Model

The regression task predicts the numerical quality score.

A Random Forest Regressor is trained using the physicochemical features. The model's predictions are compared with the actual quality scores in the test set.

The evaluation metrics are R^2 score and Mean Absolute Error.

6.4 Classification Model

The classification task predicts whether the wine is Good or Not Good.

A binary target is created using the quality threshold of 7. A Random Forest Classifier is then trained to predict the category.

The model is evaluated using Accuracy, Precision, Recall, F1-score, and a confusion matrix.

6.5 Model Persistence

The trained models are saved as Joblib files in the models/ directory.

This allows the prediction script and dashboard to load existing models without retraining them each time.

6.6 Prediction

The prediction script and dashboard accept the 11 physicochemical input values.

Based on the selected wine type, the application loads the corresponding trained models and generates:

- A predicted numerical quality score.

- A predicted Good or Not Good category.

The application can store the prediction and associated information in MongoDB.

6.7 Visualization

Matplotlib and Seaborn are used to generate evaluation charts, including regression metric comparisons, classification metric comparisons, and confusion matrices.

The Streamlit dashboard presents model performance and prediction history in an interactive format.

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7. ALGORITHMS AND EVALUATION METRICS

7.1 Random Forest Algorithm

Random Forest creates multiple decision trees and combines their predictions.

For regression, the final prediction is based on the average output of the trees:

$$
\hat{y}=\frac{1}{N}\sum_{i=1}^{N}T_i(x)
$$

Here:

- N represents the number of trees.
- $T_i(x)$ represents the prediction from tree i.
- $\hat{y}$ represents the predicted quality score.

For classification, the final class is determined from the trees' class predictions.

7.2 Accuracy

Accuracy measures the proportion of correct predictions:

$$
Accuracy=\frac{TP+TN}{TP+TN+FP+FN}
$$

7.3 Precision

Precision measures how many observations predicted as positive are actually positive:

$$
Precision=\frac{TP}{TP+FP}
$$

7.4 Recall

Recall measures how many actual positive observations are correctly identified:

$$
Recall=\frac{TP}{TP+FN}
$$

7.5 F1-score

F1-score is the harmonic mean of precision and recall:

$$
F1=2\times\frac{Precision\times Recall}{Precision+Recall}
$$

7.6 R² Score

R² measures how well a regression model explains the variation in the target variable relative to a baseline that predicts the mean.

A value closer to 1 generally indicates a better fit.

7.7 Mean Absolute Error

MAE measures the average absolute difference between predicted and actual quality scores:

$$
MAE=\frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|
$$

A lower MAE indicates smaller average prediction errors.

7.8 Confusion Matrix

A confusion matrix compares actual and predicted classes. It shows true positives, true negatives, false positives, and false negatives, helping identify the types of classification errors made by the model.

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8. MONGODB DATABASE DESIGN

8.1 Database Name

The database used in the project is:

wine_quality_db

8.2 Collections

Collection	Purpose
red_wine	Stores red wine dataset records
white_wine	Stores white wine dataset records
model_metrics	Stores model evaluation results
predictions	Stores prediction history

8.3 Data Storage

The wine collections contain the physicochemical measurements and quality values from the datasets.

The model metrics collection stores information about the dataset, algorithm, training and testing counts, evaluation metrics, classification threshold, and evaluation results.

The predictions collection stores the prediction history generated through the application.

8.4 Advantages of MongoDB Integration

- Persistent storage of dataset records.
- Centralized storage of evaluation results.
- Storage and retrieval of prediction history.
- Flexible document structure.
- Integration with Python through PyMongo.

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9. SYSTEM IMPLEMENTATION

9.1 Project Structure

WineQualityML/

├── app.py

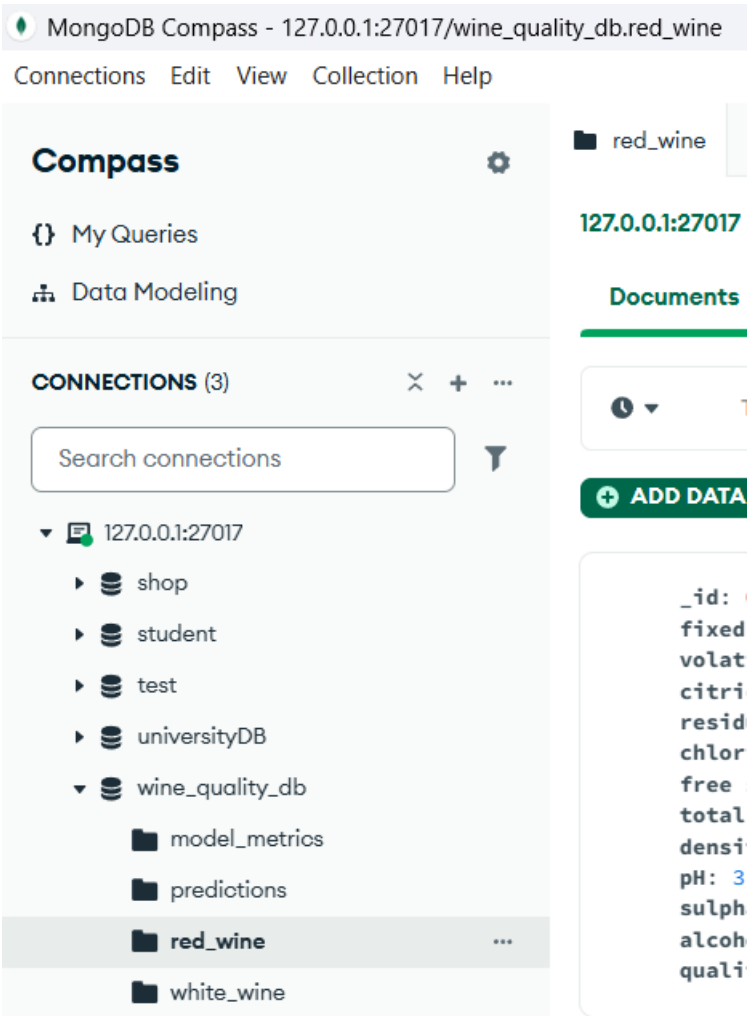
├── data/

│ ├── winequality-red.csv

```
│   ├── winequality-white.csv
│   └── winequality.names
├── models/
│   ├── red_wine_regressor.joblib
│   ├── red_wine_classifier.joblib
│   ├── white_wine_regressor.joblib
│   └── white_wine_classifier.joblib
├── results/
│   ├── regression_metrics.png
│   ├── classification_metrics.png
│   ├── confusion_matrices.png
│   ├── regression_metrics.csv
│   └── classification_metrics.csv
└── src/
    ├── load_data.py
    ├── train_models.py
    ├── visualize_results.py
    └── predict_wine.py
```

9.2 Data Loading Module

src/load_data.py loads the red and white wine CSV files and inserts their records into MongoDB.



➢ *Wine Dataset Storage in MongoDB*

9.3 Model Training Module

src/train_models.py trains the Random Forest regression and classification models, calculates evaluation metrics, saves the trained models, and stores evaluation results in MongoDB.

➢ *Model Training and Evaluation Output*

```
(.venv) C:\Users\priyanshu\WineQualityML>python src\train_models.py
MongoDB connection successful.

============================================================
Training models for red_wine
============================================================

Regression metrics:
{
  "r2_score": 0.5319,
  "mae": 0.4251
}

Classification metrics:
{
  "accuracy": 0.9031,
  "precision": 0.6538,
  "recall": 0.7234,
  "f1_score": 0.6869,
  "confusion_matrix": [
    [
      255,
      18
    ],
    [
      13,
      34
    ]
  ]
}

Models saved in: C:\Users\priyanshu\WineQualityML\models
Metrics saved in MongoDB collection: model_metrics
```

```
============================================================
Training models for white_wine
============================================================

Regression metrics:
{
  "r2_score": 0.5541,
  "mae": 0.4172
}

Classification metrics:
{
  "accuracy": 0.8765,
  "precision": 0.7227,
  "recall": 0.7577,
  "f1_score": 0.7398,
  "confusion_matrix": [
    [
      687,
      66
    ],
    [
      55,
      172
    ]
  ]
}

Models saved in: C:\Users\priyanshu\WineQualityML\models
Metrics saved in MongoDB collection: model_metrics
```

9.4 Visualization Module

src/visualize_results.py generates charts and CSV files containing the evaluation results.

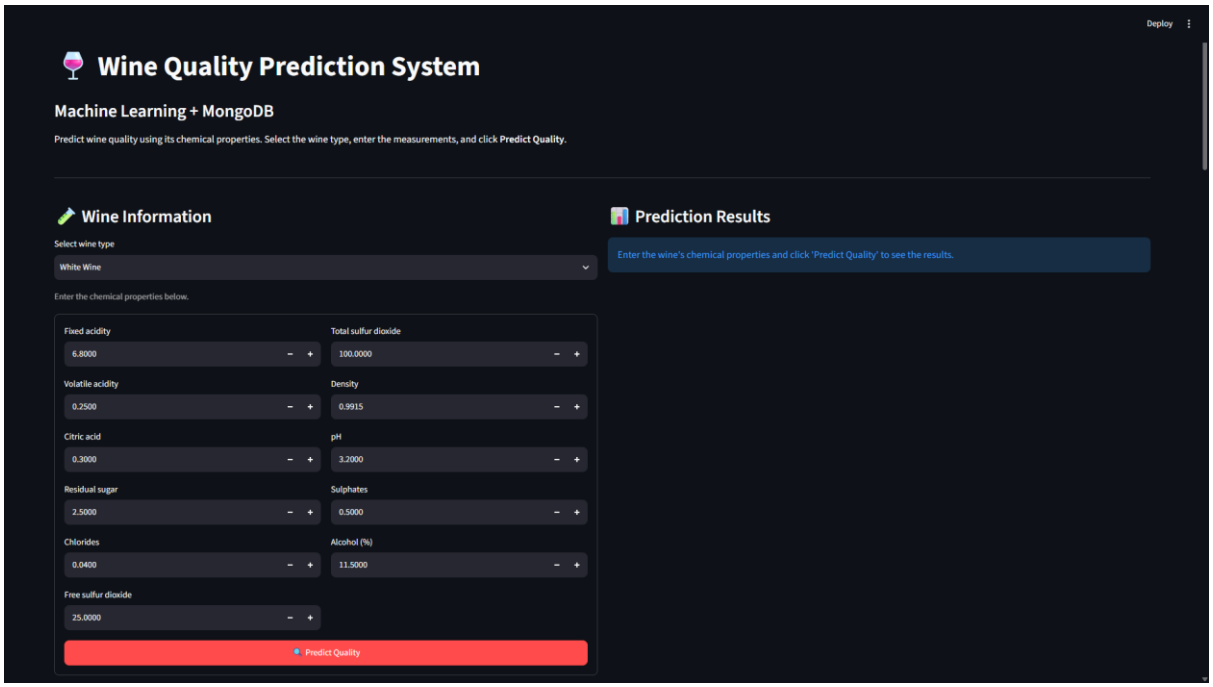
9.5 Prediction Module

src/predict_wine.py accepts input values through the terminal, loads the relevant trained models, generates predictions, and stores prediction records in MongoDB.

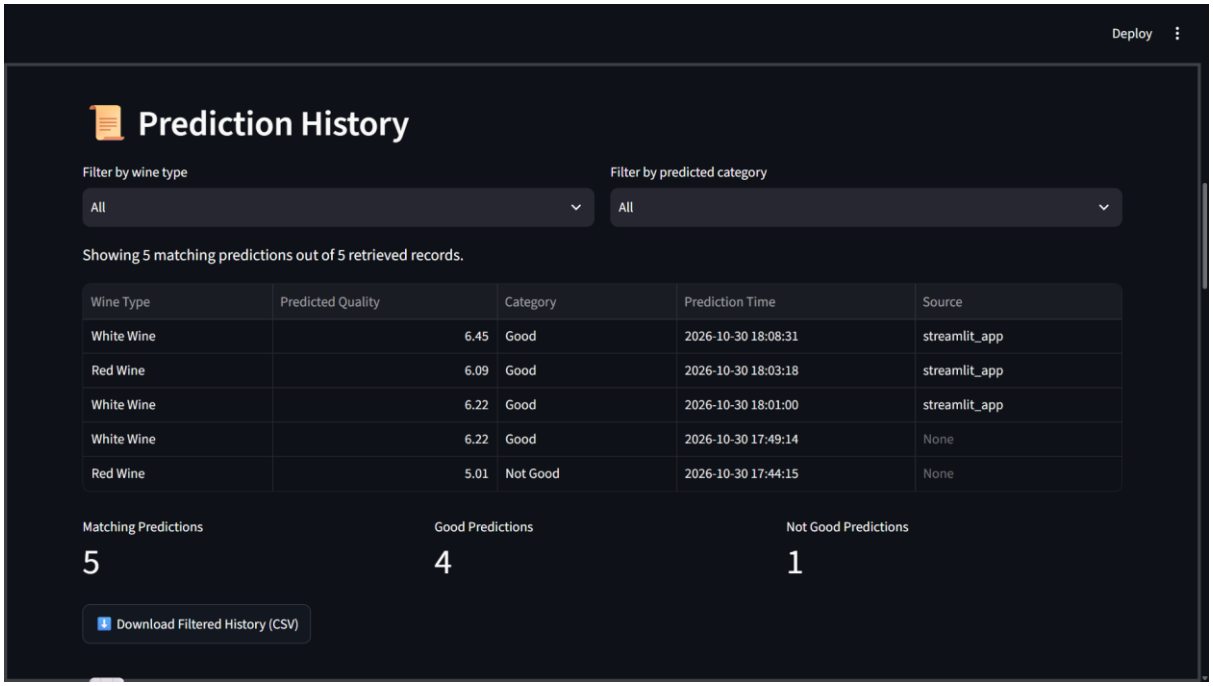
9.6 Streamlit Dashboard

app.py provides an interactive dashboard with prediction inputs, model evaluation results, visualizations, and prediction history.

The dashboard also allows users to filter historical predictions and download the displayed history as a CSV file.



➢ *Main Dashboard Interface*



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10. RESULTS AND PERFORMANCE EVALUATION

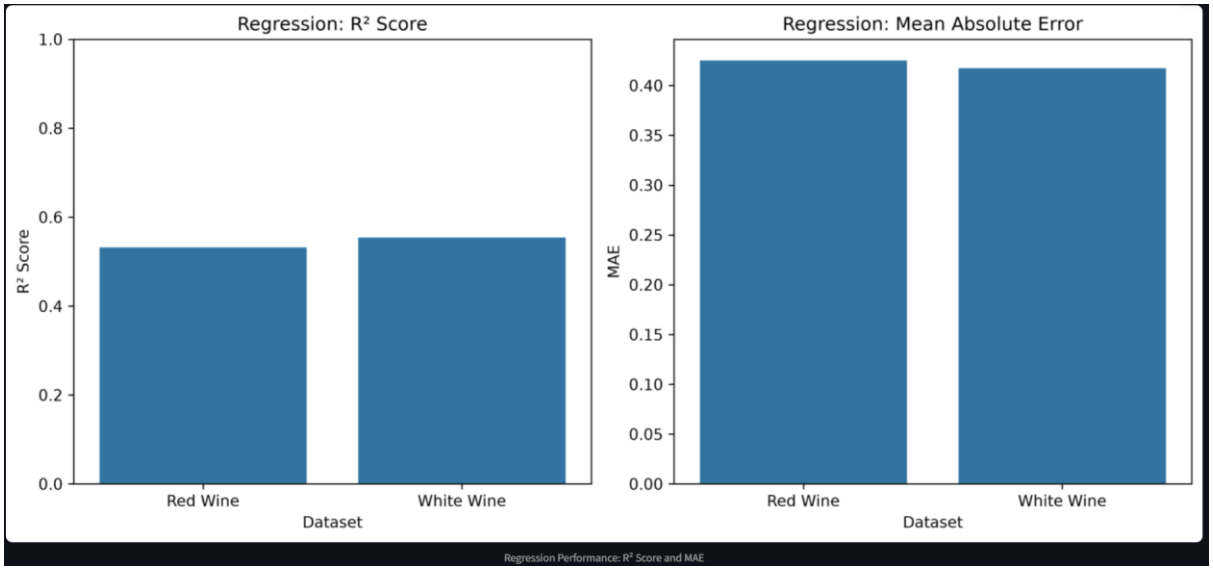
The trained models were evaluated on their respective test sets. The results below are from the current implementation.

10.1 Regression Results

Dataset	R^2 Score	MAE
Red wine	0.5319	0.4251
White wine	0.5541	0.4172

The white wine regression model achieved an R^2 score of 0.5541 and an MAE of 0.4172. The red wine regression model achieved an R^2 score of 0.5319 and an MAE of 0.4251.

These results indicate that both models learn meaningful patterns from the input features, although prediction errors remain.

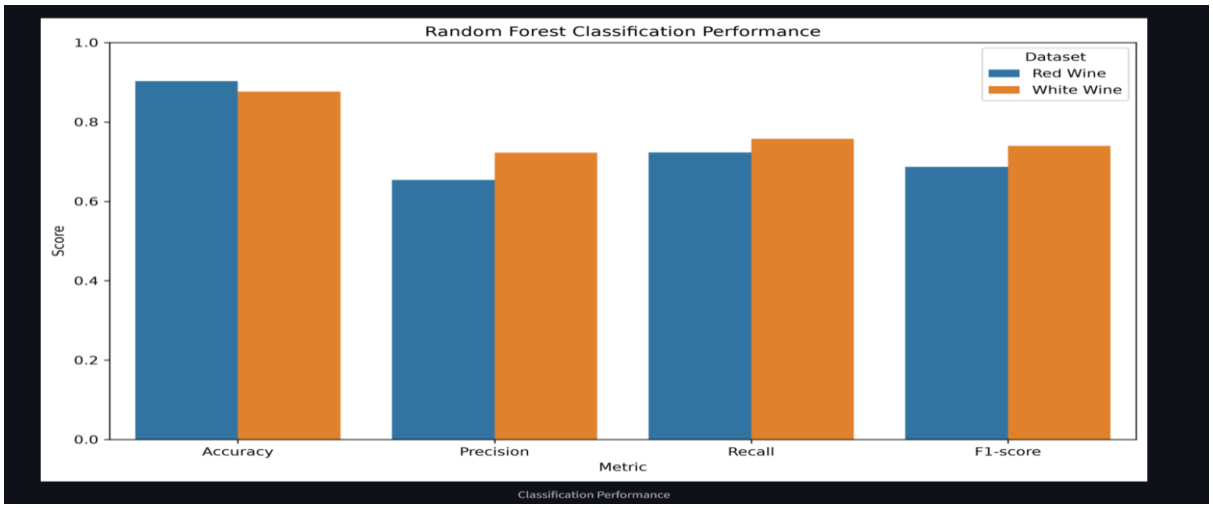


10.2 Classification Results

Dataset	Accuracy	Precision	Recall	F1-score
Red wine	90.31%	65.38%	72.34%	68.69%
White wine	87.65%	72.27%	75.77%	73.98%

The red wine classifier achieved higher overall accuracy. The white wine classifier achieved higher precision, recall, and F1-score.

Accuracy alone does not fully describe classification performance. Precision and recall are particularly useful for understanding how the model identifies wines in the Good category.



10.3 Confusion Matrices

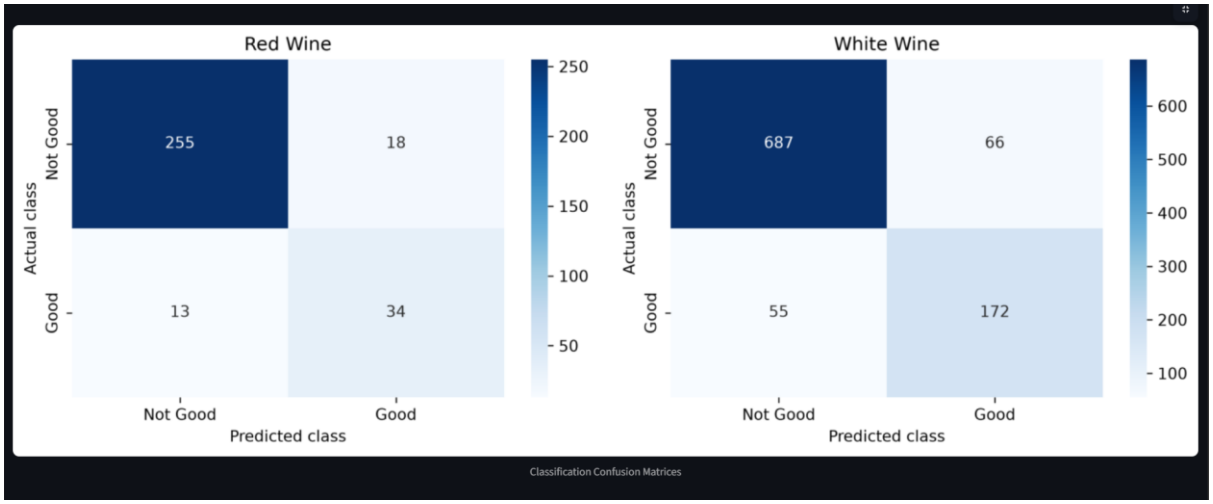
Red Wine Classification

Actual / Predicted	Not Good	Good
Not Good	255	18
Good	13	34

White Wine Classification

Actual / Predicted	Not Good	Good
Not Good	687	66
Good	55	172

The matrices show that the models correctly classify many examples in both categories but still produce false positives and false negatives.



10.4 Interpretation

The red wine classifier performs well in terms of overall accuracy, while the white wine classifier achieves a higher F1-score.

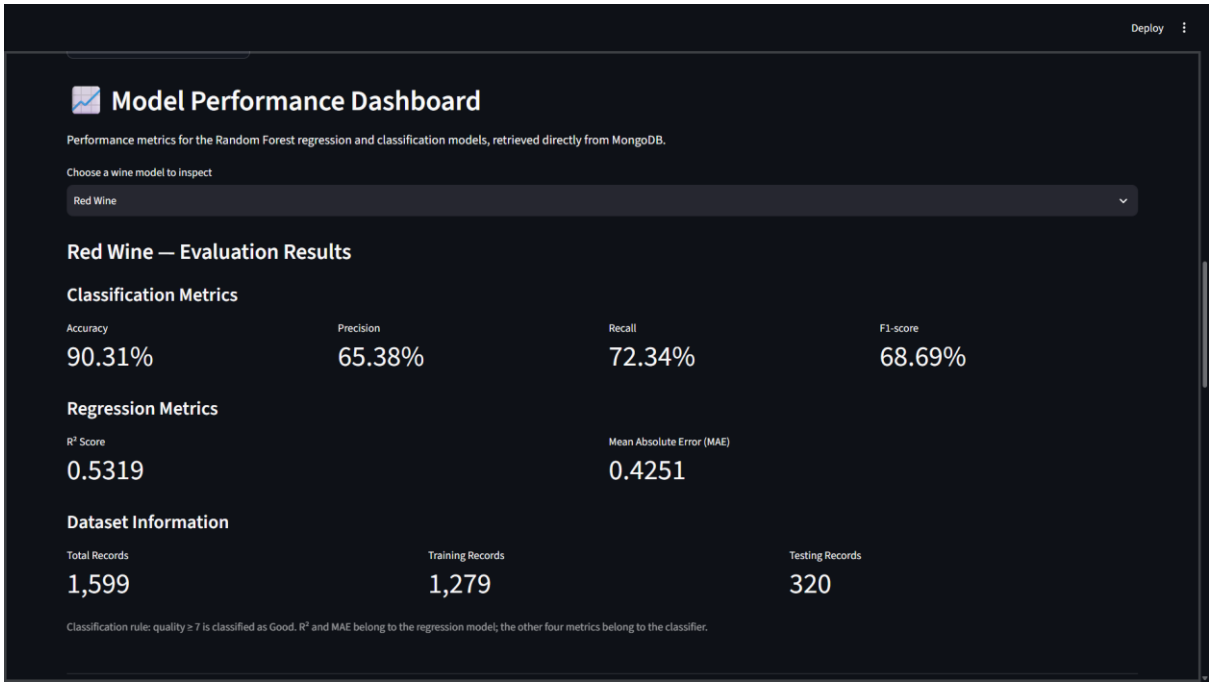
The regression results show that the models capture some of the relationships between physicochemical properties and quality scores. However, the R^2 values also indicate that substantial variation remains unexplained.

The results should be interpreted in the context of the test data and the selected classification threshold.

10.5 Dashboard Results

The Streamlit dashboard provides a user interface for entering physicochemical measurements and obtaining predictions. It also displays evaluation metrics, visualizations, and stored prediction history.

MongoDB supports persistence of evaluation metrics and prediction records.





11. LIMITATIONS

The current implementation has several limitations:

1. Model performance depends on the quality and representativeness of the training data.
2. The binary classification threshold of 7 is a project-defined rule and may not suit every application.
3. Classification accuracy can be affected by the distribution of Good and Not Good examples.
4. Regression models cannot eliminate uncertainty in subjective or complex quality assessments.
5. The current application uses a local MongoDB instance.
6. The models have been evaluated using the implemented train-test split; further validation could provide additional evidence of generalization.
7. The system has not been established as a replacement for laboratory analysis or expert evaluation.

12. FUTURE SCOPE

The project can be improved in several ways:

- Compare Random Forest with other algorithms such as Gradient Boosting and Support Vector Machines.
- Perform systematic hyperparameter tuning.
- Use cross-validation to obtain more robust performance estimates.
- Add feature importance charts to explain which chemical properties influence predictions.
- Improve handling of class imbalance if appropriate.
- Add more detailed exploratory data analysis.
- Deploy the dashboard to a suitable hosting platform.
- Use a managed database for a deployed application.
- Add model versioning and prediction monitoring.
- Improve input validation and provide clearer explanations of prediction uncertainty.

13. CONCLUSION

This project demonstrates the application of machine learning to wine quality prediction using physicochemical properties of red and white wine.

Random Forest regression models were used to predict numerical quality scores, while Random Forest classification models categorized wines as Good or Not Good. The models were evaluated using R^2 score, MAE, Accuracy, Precision, Recall, F1-score, and confusion matrices.

The project integrates Python-based machine learning with MongoDB for data storage and Streamlit for interactive visualization. The resulting application supports predictions, model performance analysis, and prediction history.

The current results demonstrate the feasibility of building an integrated prediction system. Further experimentation, validation, and model improvement could help increase predictive performance and reliability.

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14. REFERENCES

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End of Report